

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 37 | Date: 11-03-2026

Total Questions: 11 | Answer all questions.

Section 1: MCQ (6 Questions - 12 Marks)

Q1. [2 marks] (Medium)

Here are the 10 unique exam questions based on the provided content:

Q2. [2 marks] (Easy)

Which of the following is NOT explicitly listed as a core design problem for Lexi?

Options:

A) Document Structure B) Spelling Checking C) Database Connectivity D) User Operations

Q4. [2 marks] (Easy)

The internal representation of Lexi's document structure is primarily built using which design pattern?

Options:

A) Iterator Pattern B) Observer Pattern C) Composite Pattern D) Decorator Pattern

Q6. [2 marks] (Easy)

What is the name of the common abstract class that serves as the base for all elements (primitive and structural) in Lexi's document structure representation?

Options:

A) Element B) Component C) Glyph D) Node

Q9. [2 marks] (Medium)

Which design problem in Lexi explicitly mentions "line-breaking" as part of its concern?

Options:

A) Document Structure B) Formatting C) Embellishing the UI D) User Operations

Q11. [2 marks] (Hard)

Why is the uniform treatment of individual objects and compositions significant in Lexi's document structure?

Options:

A) It simplifies the user interface for end-users. B) It reduces the memory footprint of the document. C) It eliminates the need for complex conditional logic when interacting with document elements. D) It speeds up the rendering of graphics.

Section 2: Short Answer (5 Questions - 25 Marks)

Q3. [5 marks] (Medium)

Explain the primary problem that the "Document Structure" design aspect in Lexi aims to solve.

Q5. [5 marks] (Medium)

Describe how the Composite Pattern allows clients to treat individual objects and compositions of objects uniformly in Lexi's document structure.

Q7. [5 marks] (Medium)

How does the Strategy Pattern address the problem of formatting in Lexi, specifically regarding line-breaking algorithms?

Q8. [5 marks] (Hard)

Discuss the foundational flexibility provided by the recursive composition approach in Lexi's document structure design.

Q10. [5 marks] (Hard)

Compare and contrast the primary goals and mechanisms of the Composite Pattern and the Strategy Pattern as applied in Lexi's design.

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